CS 255 Business Requirements Document

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- The purpose of this project is to design an integrated system for DriverPass that provides online practice exams, manages on-the-road driving lessons, and supports student progress tracking.
- The client, **DriverPass**, wants a system that allows students to prepare more efficiently for their driving test by combining digital learning, scheduling tools, and performance analytics.
- The system must support multiple stakeholders—students, instructors, administrators, and IT staff—and enable seamless interactions between them.
- The project aims to modernize how driver-training services are delivered, ensuring reliability, accessibility, and improved student outcomes.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- DriverPass identified a significant gap in the market: over **65% of driving-test candidates fail** because they only study previous tests rather than practicing with realistic material and guided instruction.

- DriverPass wants to address this problem by offering:
  - Online practice exams
  - Course materials
  - Student progress reports
  - Automated scheduling for instructors
  - Integrated messaging
  - Self-service portal (payments, profile updates, and practice tests)

- Key system components:
  - Web-based student portal
  - Admin management portal
  - Instructor scheduling module
  - Secure database storing student info, exam attempts, lesson schedules
  - Payment integration module
  - Automated notification and confirmation system

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

The system should:

- Allow students to register, log in, schedule lessons, take online exams, and track progress.
- Enable instructors to manage availability, confirm or reject lessons, and track student performance.
- Provide administrators with oversight of accounts, schedules, system activity, and content updates.
- Ensure online tests mimic real test conditions and generate instant feedback.
- Ensure secure handling of sensitive data such as user credentials and payment information.
- Automate notifications for lesson confirmations, cancellations, payments, and exam scores.
- Support high availability, fast response times, and scalability for future expansion.
- Provide measurable outcomes such as improved student pass rates and reduced administrative workload.

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The system must operate as a **web-based and mobile-responsive** application accessible 24/7.
- Pages should load within **2–3 seconds** under normal conditions.
- System updates should occur quarterly, with critical patches applied immediately.
- Online practice exams should run without lag or delays, ensuring a smooth test experience.
- Scheduling operations should respond in real time to prevent double-booking.

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- The system must run on all major platforms (Windows, macOS, Linux) via web browser.
- Mobile browsers (Chrome, Safari, Firefox) must be fully supported for student access on phones.
- Back end should use a secure relational database (e.g., MySQL, PostgreSQL, SQL Server).
- The system must integrate with a third-party payment processor like Stripe or Square.
- The environment should support cloud deployment (AWS, Azure, or GCP) for scalability.

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- User roles must be distinguished using unique identifiers and role-based permissions.
- Input validation (case sensitivity, formatting, number ranges) must be strictly enforced to prevent errors.
- Admins should receive immediate system alerts when payment failures, instructor cancellations, or abnormal login attempts occur.
- The system must ensure 100% accuracy for scoring online exams and recording results.

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- Admins must be able to add/modify/remove course materials, test questions, and schedules without touching code.
- The system should adapt smoothly to browser or OS updates without requiring downtime.
- IT administrators require full access to logs, error reports, user management tools, and backend configuration.
- The system must support future expansion such as mobile apps, video lessons, and advanced analytics.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- Login requires unique username + password with optional MFA for admin/instructor accounts.

- All data exchanges must use **HTTPS encryption** and secure token-based authentication.
- After **5 failed login attempts**, the system should lock the account temporarily and notify the user.
- Password recovery must use email verification and secure reset links.
- Sensitive student data (payment info, test scores, personal details) must be encrypted at rest.
- Admin actions must be logged for auditing.

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- The system shall allow students to create an account and securely log in.
- The system shall allow students to schedule on-road lessons.
- The system shall validate instructor availability before confirming a lesson.
- The system shall provide online practice exams with randomized questions.
- The system shall auto-score practice exams and display results immediately.
- The system shall allow students to track progress through dashboards.
- The system shall allow instructors to view assigned lessons and update status.
- The system shall allow administrators to manage users, schedules, and course content.
- The system shall send notifications for lesson bookings, cancellations, payments, and test scores.
- The system shall integrate with a payment processor for plan purchases or lesson payments.
- The system shall store and retrieve previous exam attempts.
- The system shall restrict functionality based on user roles (student/instructor/admin).

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- User groups: Students, Instructors, Administrators, IT support.
- Students must be able to:
  - Take tests, view scores, update profile, schedule lessons, make payments.
- Instructors must be able to:
  - Manage schedules, accept/cancel lessons, track progress of assigned students.
- Admins must be able to:
  - Manage users, update test materials, monitor system performance, handle escalations.
- The UI must be accessible through:
  - Web browsers on desktop/laptop
  - Mobile browsers on phones and tablets
- The interface must be clean, intuitive, and consistent with modern UX standards.

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

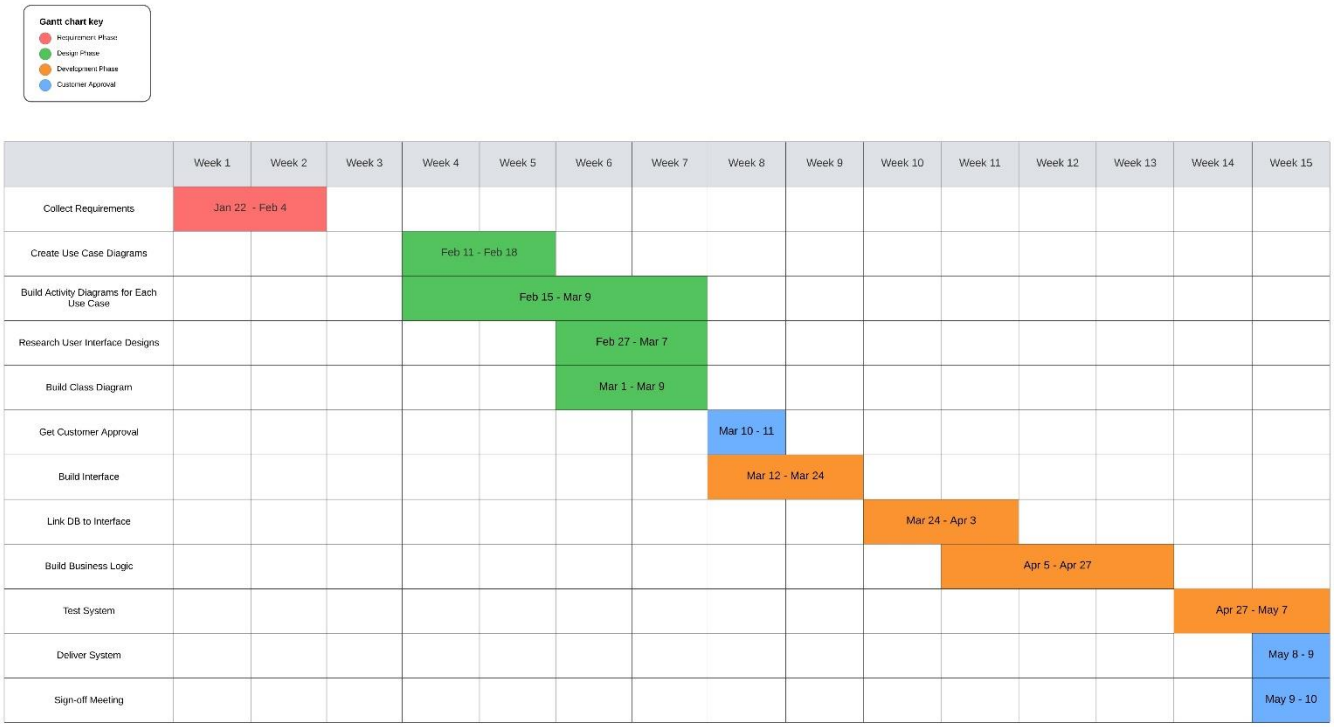
- Users have stable internet access and a modern browser.
- Instructors are trained to use digital scheduling tools.
- Students understand basic online navigation (login, dashboard use).
- DriverPass will supply all lesson materials, test content, and scheduling rules.
- Third-party services (payment gateways, email providers) operate without interruption.

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- Budget limits development to core features; advanced analytics may be added later.

- Project timeline (Jan–May) restricts scope and requires phased delivery.
- External dependencies (payment processor, email gateway) may cause downtime.
- Limited instructor availability may affect real-world scheduling despite system efficiency.
- No native mobile app in Phase 1 (web only).

Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.